

Each event SHALL be described in a section that defines the purpose of the event and the data fields of the event (if any).

### 7.14.1. Event Record

An event record is created by the node at the time the event happens. That record SHALL have the following data fields associated with it that are common to all events:

| Name            | Type        | Conformance |
|-----------------|-------------|-------------|
| Number          | event-no    | M           |
| SystemTimeStamp | sys-time-ms | O.a         |
| EpochTimeStamp  | posix-ms    | O.a         |
| Priority        | priority    | M           |
| Data            | struct      | O           |

#### 7.14.1.1. Number Field

This is an [event number](#) value that is scoped to the node. This number SHALL be monotonically increasing for the life of the node. This monotonicity guarantee SHALL be preserved across [restarts](#).

Between restarts, each event record SHALL be assigned a number that is exactly 1 greater than the last created event record on that Node.

When a node restarts, the event number MAY increase by a larger step than 1. *Rationale:* Nodes do not need to write every new value of the event number counter to permanent storage each time it is increased (e.g. to prevent flash wear due to many write operations). One example strategy to achieve reduction of non-volatile storage updates is described below:

1. Read the [counter](#) value at start-up.
2. Before processing any message, write [counter](#) + N to storage, where N is a carefully chosen number (e.g. 1000). This number N should be chosen carefully in order not to exhaust the life-time 64-bit counter space.
3. Process messages normally until the [counter](#) has a value one less than the [counter](#) in storage. When this happens, store [counter](#) + N to storage.

#### 7.14.1.2. SystemTimeStamp and EpochTimeStamp Fields

Each event record SHALL have a timestamp at the time it was created (and not when it is reported to a client). This timestamp SHALL either be [System Time in milliseconds](#) or [POSIX Time in milliseconds](#).

#### 7.14.1.3. Priority Field

Each event record has an associated [priority](#). This priority describes the usage semantics of the event.